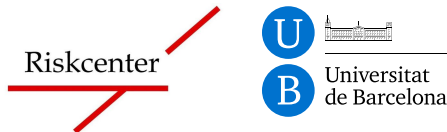


Survival analysis with actuarial applications (II)

Material created by PhD. Ramon Alemany

Departament d'Econometria, Estadística i Eco. Aplicada



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Survival analysis with actuarial applications (II)

- 1 Parametric distributions
- 2 Accelerated Failure Time Model
- 3 Proportional Hazards Regression Model

Parametric distributions

- Exponential and shifted exponential
- Weibull and shifted Weibull
- Lognormal

Parametric distributions

Exponential and shifted exponential

Exponential:

- The density function: $f(t) = \lambda \exp[-\lambda t]$
- Expectation: $E(T) = \frac{1}{\lambda}$
- Survival function: $S(t) = \exp[-\lambda t]$
- Hazard function: $\lambda(t) = \lambda$

Shifted exponential: It includes the parameter θ which can be called a guaranteed time, as the event is not going to occur before that moment in time

- Density function: $f(t) = \lambda \exp[-\lambda(t - \theta)]$
- Expectation: $E(T) = \frac{1}{\lambda} + \theta$
- Survival function: $S(t) = \exp[-\lambda(t - \theta)]$
- Hazard function: $\lambda(t) = \lambda$

Parametric distributions

Weibull and shifted Weibull

Weibull:

- Density function: $f(t) = \lambda \alpha (\lambda t)^{\alpha-1} \exp[-(\lambda t)^\alpha]$
- Expectation: $E(T) = \Gamma(1 + \frac{1}{\alpha}) / \lambda$
- Survival function: $S(t) = \exp[-(\lambda t)^\alpha]$
- Hazard function: $\lambda(t) = \lambda \alpha (\lambda t)^{\alpha-1}$

Shifted Weibull: It includes the parameter θ which can be called a guaranteed time, as the event is not going to occur before that moment in time

- Density function: $f(t) = \lambda \alpha (\lambda(t - \theta))^{\alpha-1} \exp[-(\lambda(t - \theta))^\alpha]$
- Expectation: $E(T) = \theta + \Gamma(1 + \frac{1}{\alpha}) / \lambda$
- Survival function: $S(t) = \exp[-(\lambda(t - \theta))^\alpha]$
- Hazard function: $\lambda(t) = \lambda \alpha (\lambda(t - \theta))^{\alpha-1}$

Parametric distributions

Lognormal

Lognormal

- Density function: $f(t) = \frac{1}{t\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2}\left(\frac{\ln[t]-\mu}{\sigma}\right)^2\right]$
- Expectation: $E(T) = \exp\left[\mu + \frac{\sigma^2}{2}\right]$
- Survival function: $S(t) = 1 - \Phi\left(\frac{\ln[t]-\mu}{\sigma}\right)$
where Φ is the distribution function of a $\mathcal{N}(\mu, \sigma)$
- Hazard function: $\lambda(t) = \frac{f(t)}{S(t)}$

Parametric distributions

Empirical fitting

There exists different graphical representations that can help us to select the most suitable parametric model:

- The hazard function or its log with respect to t or $\ln(t)$.
- The cumulative hazard function or its log with respect to t or $\ln(t)$.
- The survival function or another transformation with respect to t or $\ln(t)$.

For example:

- For the exponential model, we have that:

$$\ln[S(t)] = -\lambda t \text{ (linear relationship between } \ln[S(t)] \text{ and } t)$$

- For the Weibull model, we have that:

$$\ln[-\ln[S(t)]] = \ln[\Lambda(t)] = \alpha \ln[t] + \alpha \ln[\lambda]$$

Parametric distributions

Fitting a model

1. We calculate the empirical survival function.

$$\hat{S}(t) = P(T > t) = \frac{1}{n} \sum_j 1[(t_j - t) > 0] =$$

$$1 - F(t) = 1 - \frac{1}{n} \sum_j 1[(t_j - t) \leq 0]$$

2. To do the corresponding graphical representations.

Parametric distributions

Example: Exponential

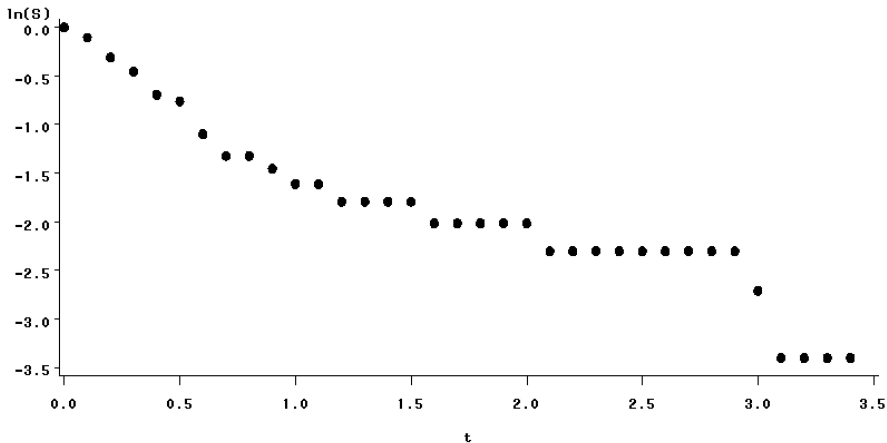
Example: Exponential ($\lambda = 1,5$)

OBS	t	OBS	t	OBS	t
1	0,0686	11	0,1155	21	0,2323
2	0,1532	12	0,3774	22	0,9678
3	3,0349	13	0,1518	23	0,4800
4	0,5533	14	0,0210	24	0,1773
5	0,2341	15	0,3648	25	0,5284
6	0,1136	16	1,1878	26	0,8009
7	0,5764	17	0,2941	27	0,0494
8	1,5452	18	0,3359	28	0,3653
9	2,9289	19	0,6208	29	2,0658
10	3,4099	20	0,5951	30	0,6465

Parametric distributions

Example: Exponential

Example: Exponential ($\lambda = 1,5$)



Parametric distributions

Example: Weibull

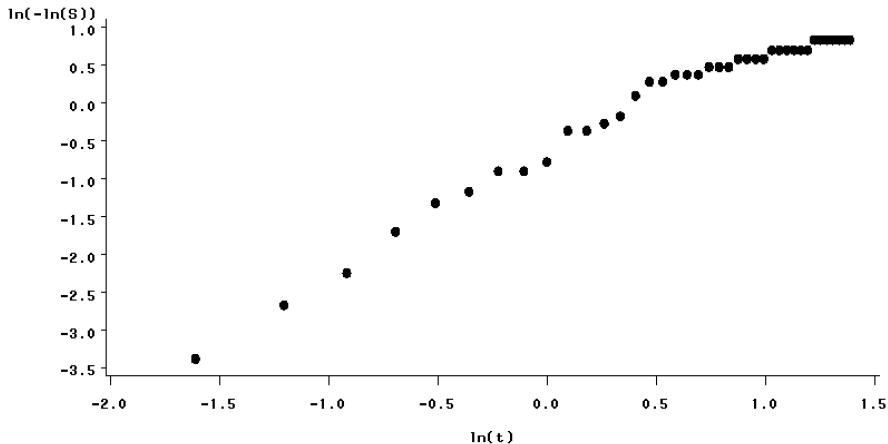
Example: Weibull ($\lambda = 0,5$, $\alpha = 1,5$)

OBS	t	OBS	t	OBS	t
1	0,3485	11	0,4934	21	0,7860
2	0,5956	12	1,0862	22	2,0352
3	4,3603	13	0,5920	23	1,2752
4	1,4019	14	0,1586	24	0,6565
5	0,7902	15	1,0620	25	1,3596
6	0,4878	16	2,3329	26	1,7939
7	1,4406	17	0,9199	27	0,2801
8	2,7801	18	1,0051	28	1,0630
9	4,2582	19	1,5137	29	3,3739
10	4,7124	20	1,4717	30	1,5552

Parametric distributions

Example: Weibull

Example: Weibull ($\lambda = 0,5$, $\alpha = 1,5$)



Introduction to regression analysis with survival data

Introduction

- We work with "heterogeneous" populations.
- We define $t > 0$ the event time and $Z = (z_1, \dots, z_p)'$ the vector of explanatory variables.
- Characteristics of the explanatory variables:
 - Quantitatives
 - Qualitatives (binary variables)
 - Time dependent variables

Accelerated Failure Time Model

Introduction

- Let $y = \ln(t)$, then

$$y = \ln(t) = \mu + \gamma Z + \sigma w$$

where $\gamma' = (\gamma_1, \dots, \gamma_p)$ is the vector of regression parameters and w is a random error.

- Different distributions have been assumed for the time to event as, for example, the Weibull or the Exponential.

Accelerated Failure Time Model

Properties

In general, the former regression model is called “Accelerated Failure Time Model (AFT)” or (“*Modelo de Tiempo de Fallo Acelerado (in Spanish)*”).

Properties:

- Let $S_0(t)$ be the survival function of $T = \exp(Y)$ when the explanatory variables in Z are equal to zero, then $S_0(t)$ is the survival function of $\exp(\mu + \sigma w)$ and, therefore:

$$P(T > t \mid Z = z) = P(Y > \ln(t) \mid Z = z) =$$

$$= P(\mu + \sigma w > \ln(t) - \gamma'Z \mid Z = z) =$$

$$= P(\exp(\mu + \sigma w) > t \exp(-\gamma'Z) \mid Z = z) = S_0(t \exp(-\gamma'z))$$

Accelerated Failure Time Model

Properties

- The effect of the explanatory variables on the original time is a change of scale given by the factor $\exp(-\gamma'Z)$.
- Depending on the sign of $\gamma'Z$ the time is accelerated (negative sign) or decelerated (positive sign).
- The hazard function of the AFT model is:

$$\lambda(t|Z) = \lambda_0(t \exp(-\gamma'Z)) \exp(-\gamma'Z)$$

Accelerated Failure Time Model

Examples with SAS

*AFM with SAS;

*Exponential;

```
proc lifereg data=data;  
model spell*event(0)=logwage ui age / distribution=exponential;  
run;
```

*Weibull;

```
proc lifereg data=data;  
model spell*event(0)=logwage ui age / distribution=weibull;  
run;
```

Accelerated Failure Time Model

Examples with SAS

The LIFEREG Procedure

Model Information

Data Set	WORK.DATA
Dependent Variable	Log(spell)
Censoring Variable	event
Censoring Value(s)	0
Number of Observations	3343
Noncensored Values	1073
Right Censored Values	2270
Left Censored Values	0
Interval Censored Values	0
Number of Parameters	4
Name of Distribution	Exponential
Log Likelihood	-2837.863053
Number of Observations Read	3343
Number of Observations Used	3343

Type III Analysis of Effects

Effect	DF	Chi-Square	Pr > ChiSq
logwage	1	71.7617	<.0001
ui	1	295.4152	<.0001
age	1	16.3808	<.0001

Fit Statistics

-2 Log Likelihood	5675.726
AIC (smaller is better)	5683.726
AICC (smaller is better)	5683.738
BIC (smaller is better)	5708.185

Analysis of Maximum Likelihood Parameter Estimates

Parameter	DF	Estimate	Standard Error	95% Confidence Limits	Chi-Square	Pr > ChiSq
Intercept	1	4.6426	0.3084	4.0381 5.2471	226.61	<.0001
logwage	1	-0.4810	0.0568	-0.5922 -0.3697	71.76	<.0001
ui	1	1.0775	0.0627	0.9546 1.2003	295.42	<.0001
age	1	0.0126	0.0031	0.0065 0.0188	16.38	<.0001
Scale	0	1.0000	0.0000	1.0000 1.0000		
Weibull Shape	0	1.0000	0.0000	1.0000 1.0000		

Fit Statistics (Unlogged Response)

-2 Log Likelihood	8167.507
Exponential AIC (smaller is better)	8175.507
Exponential AICC (smaller is better)	8175.519
Exponential BIC (smaller is better)	8199.965

Lagrange Multiplier Statistics

Parameter	Chi-Square	Pr > ChiSq
Scale	9.6527	0.0019

Algorithm converged.

Accelerated Failure Time Model

Examples with SAS

The LIFEREG Procedure

Model Information

Data Set	WORK.DATA
Dependent Variable	Log(spell)
Censoring Variable	event
Censoring Value(s)	0
Number of Observations	3343
Noncensored Values	1073
Right Censored Values	2270
Left Censored Values	0
Interval Censored Values	0
Number of Parameters	5
Name of Distribution	Weibull
Log Likelihood	-2833.568037

Number of Observations Read	3343
Number of Observations Used	3343

Fit Statistics

-2 Log Likelihood	5667.136
AIC (smaller is better)	5677.136
AICC (smaller is better)	5677.154
BIC (smaller is better)	5707.709

Fit Statistics (Unlogged Response)

-2 Log Likelihood	8158.917
Weibull AIC (smaller is better)	8168.917
Weibull AICC (smaller is better)	8168.935
Weibull BIC (smaller is better)	8199.490

Algorithm converged.

Type III Analysis of Effects

Effect	DF	Chi-Square	Pr > ChiSq
logwage	1	73.0566	<.0001
ui	1	296.2616	<.0001
age	1	18.2864	<.0001

Analysis of Maximum Likelihood Parameter Estimates

Parameter	DF	Estimate	Standard Error	95% Confidence Limits	Chi-Square	Pr > ChiSq
Intercept	1	4.4784	0.2915	3.9072 5.0496	236.11	<.0001
logwage	1	-0.4567	0.0534	-0.5614 -0.3520	73.06	<.0001
ui	1	1.0352	0.0601	0.9173 1.1531	296.26	<.0001
age	1	0.0125	0.0029	0.0068 0.0182	18.29	<.0001
Scale	1	0.9328	0.0217	0.8912 0.9764		
Weibull Shape	1	1.0720	0.0250	1.0242 1.1221		

Accelerated Failure Time Model

Examples with R

```
# AFT with R (Exponential and Weibull)

exponential <- survreg(Surv(time,event) ~ X, dist="exponential")
summary(exponential)

weibull <- survreg(Surv(time,event) ~ X, dist="weibull")
summary(weibull)
```

Accelerated Failure Time Model

Examples with R

```
> exponential <- survreg(Surv(time,event) ~ X, dist="exponential")
> summary(exponential)
```

Call:

```
survreg(formula = Surv(time, event) ~ X, dist = "exponential")
```

	Value	Std. Error	z	p
(Intercept)	4.6426	0.30841	15.05	3.27e-51
Xlogwage	-0.4810	0.05678	-8.47	2.43e-17
Xui	1.0775	0.06269	17.19	3.29e-66
Xage	0.0126	0.00312	4.05	5.18e-05

Scale fixed at 1

Exponential distribution

Loglik(model)= -4083.8 Loglik(intercept only)= -4258.4

Chisq= 349.26 on 3 degrees of freedom, p= 0

Number of Newton-Raphson Iterations: 5

n= 3343

Accelerated Failure Time Model

Examples with R

```
> weibull <- survreg(Surv(time,event) ~ X, dist="weibull")  
> summary(weibull)
```

Call:

```
survreg(formula = Surv(time, event) ~ X, dist = "weibull")
```

	Value	Std. Error	z	p
(Intercept)	4.4784	0.29145	15.37	2.78e-53
Xlogwage	-0.4567	0.05343	-8.55	1.26e-17
Xui	1.0352	0.06014	17.21	2.15e-66
Xage	0.0125	0.00292	4.28	1.90e-05
Log(scale)	-0.0695	0.02328	-2.99	2.81e-03

Scale= 0.933

Weibull distribution

Loglik(model)= -4079.5 Loglik(intercept only)= -4258.2

Chisq= 357.57 on 3 degrees of freedom, p= 0

Number of Newton-Raphson Iterations: 5

n= 3343

Proportional Hazards Regression Model (Cox)

Introduction

The Proportional Hazards Regression Model is formulated as:

$$\lambda(t|Z) = \lambda_0(t) c(\beta'Z)$$

where $\lambda_0(t)$ is the “baseline” hazard function and $c(\beta'Z)$ is a function of the explanatory variables and takes non negative values.

For example, in the Cox Model (Cox, 1972) we have that $c(\beta'Z) = \exp(\beta'Z)$ and therefore, the model is:

$$\lambda(t|Z) = \lambda_0(t) \exp(\beta'Z)$$

Proportional Hazards Regression Model (Cox)

Properties

Properties:

- For fixed covariates at time 0, the hazard functions of two individuals with different values for Z are proportional:

$$\frac{\lambda(t|Z^1)}{\lambda(t|Z^2)} = \frac{\lambda_0(t) c(\beta'Z^1)}{\lambda_0(t) c(\beta'Z^2)} = \frac{c(\beta'Z^1)}{c(\beta'Z^2)} = \text{Constant and independent of } t$$

- In terms of the survival function:

$$\lambda(t|Z) = \lambda_0(t) c(\beta'Z) \implies S(t|Z) = S_0(t)^{c(\beta'Z)}$$

$$\begin{aligned} \lambda(t|Z) &= -\frac{d}{dt} \ln [S(t|Z)] \implies \lambda_0(t) c(\beta'Z) = -\frac{d}{dt} \ln [S(t|Z)] \implies \\ \implies c(\beta'Z) \int \lambda_0(t) dt &= -\ln [S(t|Z)] \implies c(\beta'Z) \left[-\int \lambda_0(t) dt \right] = \ln [S(t|Z)] \implies \\ \implies -\int \lambda_0(t) dt &= \frac{1}{c(\beta'Z)} \ln [S(t|Z)] \implies -\int \lambda_0(t) dt = \ln \left[S(t|Z)^{\frac{1}{c(\beta'Z)}} \right] \implies \\ \implies \exp \left[-\int \lambda_0(t) dt \right] &= S(t|Z)^{\frac{1}{c(\beta'Z)}} \implies S_0(t)^{c(\beta'Z)} = S(t|Z) \end{aligned}$$

Proportional Hazards Regression Model (Cox)

Examples with SAS

```
*Semiparametric model;  
  
*Cox proportional hazard model;  
  
proc phreg data=data;  
model spell*event(0)=logwage ui age;  
run;
```

Proportional Hazards Regression Model (Cox)

Examples with SAS

The PHREG Procedure				Model Fit Statistics						
Model Information				Criterion	Without Covariates	With Covariates				
Data Set	WORK.DATA			-2 LOG L	15963.861	15682.333				
Dependent Variable	spell			AIC	15963.861	15688.333				
Censoring Variable	event			SBC	15963.861	15703.268				
Censoring Value(s)	0									
Ties Handling	BRESLOW			Testing Global Null Hypothesis: BETA=0						
Number of Observations Read	3343			Test	Chi-Square	DF	Pr > ChiSq			
Number of Observations Used	3343			Likelihood Ratio	281.5275	3	<.0001			
				Score	300.0015	3	<.0001			
				Wald	286.3086	3	<.0001			
Summary of the Number of Event and Censored Value										
				The PHREG Procedure						
Total	Event	Censored	Percent Censored	Analysis of Maximum Likelihood Estimates						
3343	1073	2270	67.90	Parameter	DF	Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
				logwage	1	0.46155	0.05719	65.1321	<.0001	1.587
				ui	1	-0.97958	0.06398	234.4246	<.0001	0.375
				age	1	-0.01085	0.00313	12.0029	0.0005	0.989
Convergence criterion (GCONV=1E-8) satisfied.										

Proportional Hazards Regression Model (Cox)

Examples with R

```
# Cox proportional hazard model
> coxph <- coxph(Surv(time,event) ~ X, method="breslow")
> summary(coxph)
```

Call:

```
coxph(formula = Surv(time, event) ~ X, method = "breslow")
```

n= 3343, number of events= 1073

	coef	exp(coef)	se(coef)	z	Pr(> z)	
Xlogwage	0.461553	1.586535	0.057190	8.070	6.66e-16	***
Xui	-0.979578	0.375470	0.063979	-15.311	< 2e-16	***
Xage	-0.010850	0.989209	0.003132	-3.465	0.000531	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
Xlogwage	1.5865	0.6303	1.4183	1.7747
Xui	0.3755	2.6633	0.3312	0.4256
Xage	0.9892	1.0109	0.9832	0.9953

Concordance= 0.693 (se = 0.011)

Rsquare= 0.081 (max possible= 0.992)

Likelihood ratio test= 281.5 on 3 df, p=0

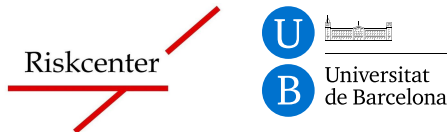
Wald test = 286.3 on 3 df, p=0

Score (logrank) test = 300 on 3 df, p=0

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